

1. Experiment Hypothesis

Experiment background

Automation tools involving artificial intelligence have been prevalent in applications of different types of industries around the world in the last couple of years. For example, image recognition, voice recognition and language translation are popular with the advance of deep learning neural networks. Much of them are based on the same or similar learning algorithms trained with the business's users data.

However, using artificial intelligence as a long running Autonomous Agent orchestrating various tools to achieve a bigger objective and with user privacy is a novel field. It requires experimenting with strong reasoning from the Large Language Model LLM, a new type of memory storage model and a new way of integrating with tools and communicating with external systems and smart contracts.

Autonomous AI Agent is a new concept. The benchmark for this is a simple program running as a main loop continuously utilizing ChatGPT as a reasoning model and connecting to tools like Google for web search.

The core issues are:

- Lack of user privacy using ChatGPT model with user's sensitive data leaking to ChatGPT for training and showing up in other users' result;
- Limited choice of Large Language Models;
- Limited choice of tools;
- Lack integration with Smart Contracts;
- Lack of data ownership;
- Absence of standard software tools or frameworks to enable integration of Autonomous AI Agent with external tools. It requires significant technical or interoperability challenges to be overcome.

A true versatile Autonomous AI Agent solution does not currently exist.

The hypothesis of the experiment is that ArcMind (Autonomous AI Agent) can be developed to:

- Understand the goal set out by the user;
- Achieve a goal set out by the user by breaking it down into multiple subtasks in background using the available tools and the reasoning model without user's intervention;
- Conduct each subtask with 100% user privacy and in safe way;
- Interoperate with various types of tools including Blockchain smart contracts;
- Use different type of reasoning models including open source Large Language Models for reasoning;
- Have a standard way to integrate with external tools;
- Save users' data in new type of database in temporary or permanent basis for efficiently processing of tasks;
- Replace bespoke programming or coding of business rules with Autonomous AI Agent.

2. New Knowledge the Core Activity Intended to Produce

New knowledge will be generated in the form of an Autonomous AI Agent completing a goal through finishing a number of subtasks planned by itself and through self-criticism and retrospective process to assess its own performance and reason about it.

Specific new knowledge:

- Understanding of how to develop an algorithm that can understand user's goal, break it down into subtasks and finish each task with self-assessment, repeat the process until the goal is completed;
- Understanding of how to develop the system with user privacy and ownership;
- Understanding of what levels of human feedback is necessary to complete a goal and improve its efficiency;
- Understanding of how to develop API that can help interoperate with external systems;

- Understanding of how to integrate with smart contract;
- Understanding of how effective the agent would be to complete user's goal comparing with human;
- Understanding of what type of datastore the Agent need for temporary and permanent basis.

3. Sources Investigated and Why the Outcome Could Not Be Determined in Advance

Basis selected in the application:

- Experts in the field provided advice that there wasn't a solution that could be applied.
- There wasn't a way to adapt solutions from other companies in, and out of, Australia.

Autonomous AI Agent is different from the single purpose AI tools that have existed for years. It's a novel idea. Most of the study are conceptual. Smart contract integration is even more novel.

We have conducted extensive research to investigate the known knowledge:

- Formal literature review on Autonomous AI Agent technology, including:
- GitHub code repositories like AutoGPT and LangChain;
- Technical journal papers;
- Existing industry standards;
- Engagement with Dfinity Foundation for blockchain integration.

4. The Experiment and How It Tested the Hypothesis

Testing and activities undertaken to prove the hypothesis included:

- Design and writing of Chain Of Thoughts algorithm to break down user's goal into multiple subtasks with integration to Large Language Model LLM;
- Writing testing framework to determine the accuracy and correctness of the algorithm;
- Design and writing of Web Browsing and Google Search Task Executor with 100% user privacy and safety;
- Design and writing of Smart Contract Task Executor with 100% user privacy and safety;
- Design, writing and testing of new type of database suitable for Autonomous AI Agent in temporary and permanent basis for efficient processing of tasks with vector, multiple dimensions and semantic similarity search;
- Writing of Web Chat Interface to support testing the whole system trial to determine the system performance from end to end perspective;
- Designing and writing of standard API to integrate with Autonomous AI Agent with existing software systems to overcome interoperability challenges and two ways handshaking protocol;
- Writing testing framework to compare correctness of implementing business rules in Autonomous AI Agent with goal-based approach versus bespoke programming by integration with BeamFi streaming payment smart contract as test trial;
- Testing of the whole system to ensure reasonable efficiency and accuracy.

5. Evaluation of Results

Results will be evaluated against:

- Accuracy and correctness of the Chain Of Thoughts in understanding a goal, planning, breaking down into tasks and reasoning with different set of parameters and configurations;
- Efficiency, security and performance of the system;
- User privacy, data ownership, system ownership and control;
- Safety for user to adopt Autonomous AI Agent;

- Flexibility of the system to use different types of Large Language Model LLM for reasoning;
- Integration quality and performance connecting with external system;
- Programming language consideration to implement and testing the system efficiently;
- Flexibility of standard API in integrating with external systems.

6. Conclusion

The experiment as of 2023/24 was considered partially successful. While the core hypothesis of Autonomous AI agent (ArcMind) is proven, we also encountered performance issues and Large Language Model limitations in AI workflow planning. Our engineering team has implemented workarounds for those challenging problems.

Technical components proven:

- Autonomous AI Agent (ArcMind) can understand the user goal, plan sub-tasks and execute the plan using the Chain of Thoughts engine we developed with external Web tools. LLM cognitive reasoning is good enough to drive the goal planning process to reach a result without human intervention.
- ArcMind can be developed with 100% user privacy and security, by having users being the owner of their agent using NFID wallet integration running on blockchain. The ownership hierarchy allows users to control their AI agents directly. To achieve 100% privacy and safety, open source LLM model Llama can be used instead of OpenAI GPT4.
- ArcMind can interoperate with smart contracts using our BeamFi streaming payment as testing. By adding special tools and prompt engineering on supported commands, ArcMind can understand the user goal and execute the plan by calling smart contracts and observing its result. However, having a standard way to integrate with external tools was proven challenging as every tool has its own kind of input and output schema. New experiment would be needed to allow LLM to understand smart contract API or schema and generate input accordingly.
- A new type of database "ArcMind VectorDB" was designed and implemented using kd-tree structure running on blockchain. It is used for storing ArcMind's long term memory data for future goal similarity search such that existing relevant information would be extracted to complement the new goal's context data, for improved efficiency. While ArcMind operates with text only, our experiments have proven that a versatile VectorDB can be developed to work on vector data type like image, audio or video.
- A simple key-value in-memory datastore can be developed in blockchain as short-term memory of ArcMind to store all the sub-tasks questions and responses from LLM and tools during goal processing.
- When performance is not a concern and the right tools are given, ArcMind can replace bespoke coding of business workflow in traditional systems. Our experiment shows that ArcMind can handle errors properly and is able to pivot to alternative solutions.

Aspects of the hypothesis not supported:

- ArcMind's Chain of thoughts process requires a lot more round-trips of LLM inferencing requests than typical AI agents. While ArcMind is safer, more secure and capable of running more sophisticated goals, its performance is suboptimal. It takes 30 seconds to create a plan for a goal. The issue is made worse with blockchain's security features when making HTTP outcalls. It requires consensus from 3 replicas to complete a HTTP request, causing 7 seconds latency delays when making LLM inferencing requests or web requests. Also, it works only on IPV6 supported websites due to limitations on blockchain. To work around the issue, we developed a HTTP proxy running on Google Cloud.
- LLM hosting on blockchain is not feasible as it requires more powerful GPU computation. We work around it by giving users options to use OpenAI GPT-4o model or Cloud hosted open source models Llama.
- While ArcMind can run sophisticated workflow autonomously, it doesn't mean the result is the best or the way it was solved is the most efficient path. As with any type of problem, there are many different ways to solve a problem. New experiment is required on Problem/Results search optimization during the model inferencing phase.
- From a performance and efficiency perspective, the current version of ArcMind does not meet our criteria expectations. New experiment is required to research on integrating chain of thoughts process in a machine learning model.